Convergence Results for Gradient Descent

- Let $f: \mathbb{R}^n \to \mathbb{R}$ be convex with global minimum x^*. We minimize f using gradient descent with constant step size γ.

$$x^{k+1} = x^k - \gamma \cdot g^k, \quad \text{where} \quad g^k = \nabla f(x^k)$$

Preliminary Analysis:

- We did a preliminary analysis to bound the cumulative "error" in gradient descent iterates over K iterations:

$$\sum_{k=0}^{K-1} \left(f(x^k) - f(x^*) \right) \leq \frac{\gamma}{2} \cdot \sum_{k=0}^{K-1} \|g^k\|^2 + \frac{1}{2\gamma} \|x^0 - x^*\|^2$$

Thm: Let $f: \mathbb{R}^n \to \mathbb{R}$ convex with global minimum x^* and $\|\nabla f(x)\| \leq B$. Assume that $\|x^0 - x^*\| \leq R$ and we use gradient descent with constant step size

$$\gamma = \frac{R}{B\sqrt{K}},$$

Then, the iterates of gradient descent satisfies

$$\frac{1}{K} \sum_{k=0}^{K-1} \left(f(x^k) - f(x^*)\right) \leq \frac{RB}{\sqrt{K}}$$

- By the theorem $\min_{k=0,\dots,K-1} f(x^k) - f(x^*) \leq \frac{1}{K} \cdot \sum_{k=0}^{K-1} \left(f(x^k) - f(x^*)\right) \leq \frac{RB}{\sqrt{K}}$.

- If we run gradient descent for K iterations, the best solution that we reach along the way will yield an objective value that deviates from optimal obj. value by at most $RB/\sqrt{K}$.

- Asking a similar question conversely, let's say we want to get a solution with error ε. How many iterations should we run gradient descent for?

Set $\varepsilon = \dfrac{RB}{\sqrt{K}} \Rightarrow K = \left(\dfrac{RB}{\varepsilon}\right)^2$

- If $\varepsilon = 0{,}1$, $K =$ constant $\cdot$ 100

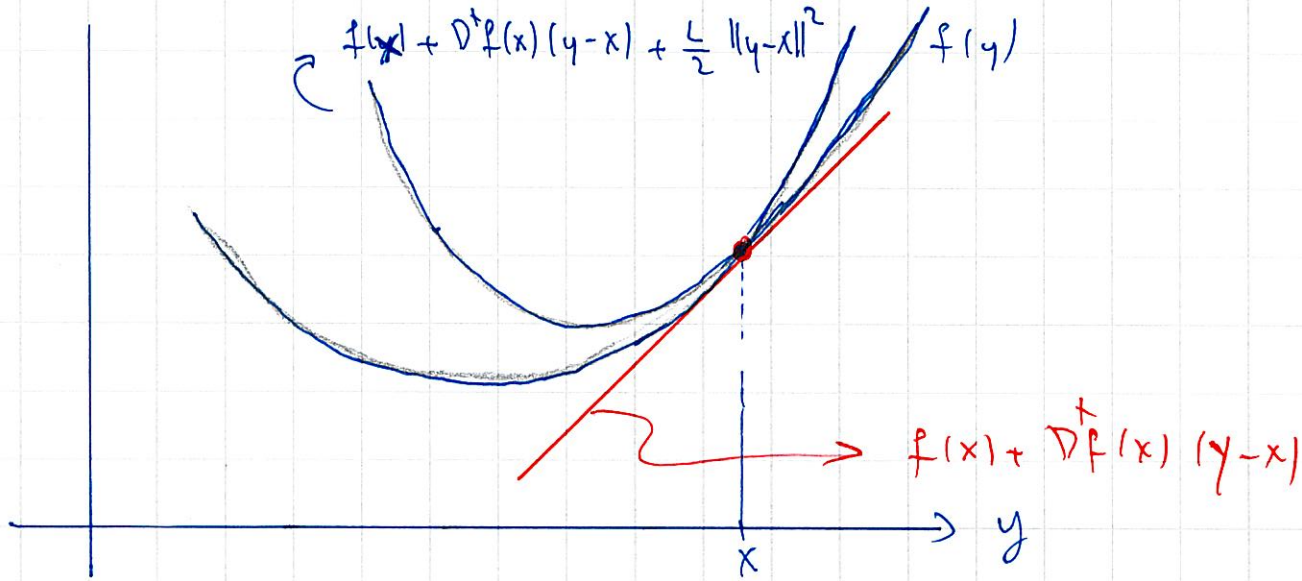
If $\varepsilon = 0{,}001$ $K =$ constant 1000000

The # of itns increase fast as we want more precision.

Convergence of Gradient Descent for Smooth Convex Functions

Definition: Let $f: \mathbb{R}^n \to \mathbb{R}$ be a convex function. We say that f is smooth with parameter L if

$$f(y) \leq f(x) + \nabla f^t(x)\,(y-x) + \frac{L}{2}\cdot\|y-x\|^2 \quad \text{for all } x, y \in \mathbb{R}^n$$



Remember also for convex f, we have $f(y) \geq f(x) + \nabla f^t(x)\cdot(y-x)$

- Intuitively, a smooth convex function lies below a local quadratic approximation, above a local linear approximation.

We need two preliminary results for analyzing gradient descent for convex smooth functions

Result 1: Let $f : \mathbb{R}^n \to \mathbb{R}$ be smooth and convex and $x^0, x^1, x^2 \ldots$ are generated by gradient descent with constant step size γ

$$x^{k+1} = x^k - \gamma g^k \quad , \quad g^k = \nabla f(x^k).$$

Then, $$f(x^{k+1}) \leq f(x^k) - \gamma \cdot \left(1 - \frac{\gamma L}{2}\right) \cdot \|g^k\|^2$$

Pf: Because f is smooth with parameter L,

$$f(x^{k+1}) \leq f(x^k) + \underbrace{\nabla f(x^k)}_{g^k}{}^t \cdot \underbrace{(x^{k+1} - x^k)}_{-\gamma g^k} + \frac{L}{2} \|\underbrace{x^{k+1} - x^k}_{\gamma g^k}\|^2$$

$$= f(x^k) - \gamma \cancel{(g^k)^t g^k} + \frac{L}{2} \gamma^2 \cancel{\|g^k\|^2}$$

$$= f(x^k) - \gamma \left(1 - \frac{L\gamma}{2}\right) \|g^k\|^2 \qquad \text{done!}$$

- Thus, with $\gamma = 1/L$, $$f(x^{k+1}) \leq f(x^k) - \frac{1}{2L} \|g^k\|^2$$

Result 2: this is our earlier preliminary analysis:

$$\sum_{k=0}^{K-1} \left(f(x^k) - f(x^*) \right) \leq \frac{\gamma}{2} \sum_{k=0}^{K-1} \|g^k\|^2 + \frac{1}{2\gamma} \|x^0 - x^*\|^2$$

<u>Theorem:</u> Let $f: \mathbb{R}^n \to \mathbb{R}$ be a smooth convex function with parameter L having global minimum x^*.
Assume that $\|x^0 - x^*\| \leq R$

Choosing step size $\gamma = 1/L$, the iterates of gradient descent algorithm satisfy

$$f(x^K) - f(x^*) \leq \frac{L}{2K} \cdot R^2$$

Pf: By result 1 above, $\frac{1}{2L}\|g^k\|^2 \leq f(x^k) - f(x^{k+1})$

Add this inequality over $k=0, \dots, K-1$ $\quad \frac{1}{2L}\sum_{k=0}^{K-1}\|g^k\|^2 \leq f(x^0) - f(x^K)$ (7)

(use telescoping sums)

By result 2 with step size $\gamma = 1/L$

$$\sum_{k=0}^{K-1}\left(f(x^k) - f(x^*)\right) \leq \frac{1}{2L}\sum_{k=0}^{K-1}\|g^k\|^2 + \frac{L}{2}\|x^0 - x^*\|^2$$

$$\underset{\text{from (7)}}{\leq} f(x^0) - f(x^K) + \frac{L}{2}R^2 \quad \Longrightarrow$$

$$\cancel{f(x^0)} - f(x^*) + \sum_{k=1}^{K-1} \left(f(x^k) - f(x^*) \right) - \cancel{f(x^0)} + f(x^K) \leq \frac{L}{2} R^2 \implies$$

$$\sum_{k=1}^{K} \left(f(x^k) - f(x^*) \right) \leq \frac{L}{2} R^2. \implies$$

$$\frac{1}{K} \sum_{k=1}^{K} \left(f(x^k) - f(x^*) \right) \leq \frac{L}{2K} R^2$$

- By result 1: $f(x^1) \geq f(x^2) \geq f(x^3) \geq \dots \geq f(x^K)$, so
  $f(x^1) - f(x^*) \geq f(x^2) - f(x^*) \geq f(x^3) - f(x^*) \geq \dots \geq f(x^K) - f(x^*)$

- we get $f(x^K) - f(x^*) \leq \frac{1}{K} \sum_{k=1}^{K} \left(f(x^k) - f(x^*) \right) \leq \frac{L}{2K} R^2$ Done!

- How many iterations to run gradient descent for so that the iterate at the last iteration gives a function value that deviates from optimal by ε?

set $\varepsilon = \frac{L}{2K} R^2 \Rightarrow K = \frac{LR^2}{2\varepsilon}$

- If $\varepsilon = 0{,}1$ then $K = \text{Const.}\ 10$

If $\varepsilon = 0{,}001$ then $K = \text{Const}\ 1000.$

- Also, we have $0 \leq f(x^K) - f(x^*) \leq \frac{L}{2K} R^2$

Thus, $0 \leq \lim_{K\to\infty} \left(f(x^K) - f(x^*)\right) \leq \lim_{K\to\infty} \frac{L}{2K} R^2 = 0$

So $\lim_{K\to\infty} \left(f(x^K) - f(x^*)\right) = 0 \Rightarrow \lim_{K\to\infty} f(x^K) = f(x^*)$

So if we run gradient descent long, the last iterate will be arbitrarily close to optimal.